Lecture 06

Erik An

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1 Proof Methods

1.1 Direct Proof

We want to prove an implication of the form $P \Rightarrow Q$.

1.1.1 Theorem 1: Odd Integers

Statement: The square of an odd integer is odd. **Proof**: Assume x is odd. Then there exists an integer $n \in \mathbb{Z}$ such that:

$$x = 2n + 1$$

Squaring x:

$$x^2 = (2n+1)^2 = 4n^2 + 4n + 1 = 4n(n+1) + 1$$

Since $4n(n+1)$ is divisible by 2 (it is even), adding 1 makes the result odd.

$$2 \nmid x^2$$

Therefore, x^2 is odd.

1.1.2 Theorem 2: Transitivity of Divisibility

Statement: Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $b \mid c$, then $a \mid c$. **Proof**:

- If $a \mid b$, then there exists an integer m such that $b = a \cdot m$.
- If $b \mid c$, then there exists an integer n such that $c = b \cdot n$.

Substituting b:

$$c = (a \cdot m)n = a(mn)$$

Since mn is an integer, $a \mid c$.

1.1.3 Theorem 3: Parity of Polynomial

Statement: If x is an even integer, then $x^2 - 6x + 5$ is odd. **Proof**: If x is even, there exists $n \in \mathbb{Z}$ such that $x = 2n$. Substituting into the expression:

$$\begin{aligned} x^2 - 6x + 5 &= (2n)^2 - 6(2n) + 5 \\ &= 4n^2 - 12n + 5 \\ &= 4n^2 - 12n + 4 + 1 \\ &= 4(n^2 - 3n + 1) + 1 \end{aligned}$$

The term $4(n^2 - 3n + 1)$ is clearly even. Thus, the sum is odd.

1.2 Proof by Contraposition

To prove $P \Rightarrow Q$, we prove the logically equivalent statement $\neg Q \Rightarrow \neg P$.

1.2.1 Theorem 4: Oddness Implication

Statement: Let $n \in \mathbb{N}$. If $3n + 2$ is odd, then n is odd. **Proof (by Contraposition)**:

- Assume $\neg P$: Suppose n is even.
- Then there exists k such that $n = 2k$.
- Substitute into the expression:

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$$

- Since $2(3k + 1)$ is divisible by 2, $3n + 2$ is even.
- This proves $\neg P \Rightarrow \neg Q$, which is equivalent to $Q \Rightarrow P$. (Note: The labels P and Q in the source text seem swapped relative to standard notation, where the hypothesis is P. Here, the source uses $\neg Q \Rightarrow \neg P$ to prove the original statement).

Attempt at Direct Proof (for comparison): If $3n + 2$ is odd:

$$3n + 2 = 2a + 1$$

$$3n + 1 = 2a$$

This path is less clear algebraically than the contrapositive method.

1.2.2 Theorem 5: Even Square

Statement: Let $n \in \mathbb{N}$. If n^2 is even, then n is even. **Proof (by Contraposition)**:

- Assume n is odd.
- We must prove that n^2 is odd.
- This was already proven in "Theorem 1".
- Since n odd $\implies$ n^2 odd, the contrapositive (n^2 even $\implies$ n even) holds true.

1.3 Proof by Contradiction (Reductio ad absurdum)

To prove $P \Rightarrow Q$, we assume P is true and Q is false, then derive a contradiction (False).

1.3.1 Theorem 6: Prime Divisors Sum

Statement: Let $n \in \mathbb{N}$. If the sum of positive divisors of n is $n+1$, then n is prime. **Proof**:

- Assume the statement is false. That is, assume the sum of divisors is $n+1$, but n is **not** prime (composite).
- If n is composite, it has a divisor a such that $1 < a < n$ and $a \mid n$.
- The set of positive divisors includes at least $\{1, a, n\}$.
- The sum of divisors is at least $1 + a + n$.
- Since $a > 0$, $1 + a + n > n + 1$.
- This contradicts the hypothesis that the sum is exactly $n + 1$.
- Therefore, the assumption that n is composite must be wrong. n is prime.

1.3.2 Theorem 7: Irrationality of $\sqrt{2}$

Statement: $\sqrt{2}$ is irrational. **Proof**:

- Assume the contrary: $\sqrt{2}$ is rational.
- Then there exist integers $m, n \in \mathbb{N}$ such that $\sqrt{2} = \frac{m}{n}$, where the fraction is in lowest terms (i.e., m and n share no common factors).
- Squaring both sides:

$$2 = \frac{m^2}{n^2} \implies m^2 = 2n^2$$

- This implies m^2 is even. By Theorem 5, m must be even.
- Let $m = 2k$ for some integer k. Substitute back:

$$(2k)^2 = 2n^2$$

$$4k^2 = 2n^2$$

$$2k^2 = n^2$$

- This implies n^2 is even, so n must be even.

- **Contradiction**: We concluded that both m and n are even (divisible by 2). This contradicts the assumption that $\frac{m}{n}$ is in lowest terms (no common factors).
- Therefore, $\sqrt{2}$ must be irrational.

1.3.3 Theorem 8: Infinitude of Primes

Statement: There are infinitely many prime numbers. **Proof**:

- Assume there is a finite number of primes: $p_1, p_2, \ldots, p_n$.
- Consider the number $a = p_1 \cdot p_2 \cdot \cdots \cdot p_n + 1$.
- This number a is not divisible by any of the primes p_i (dividing by any p_i leaves a remainder of 1).
- Therefore, a must either be prime itself or have a prime factor not in the list $\{p_1, \ldots, p_n\}$.
- Both cases contradict the assumption that the list contains **all** primes.

1.4 Additional Problem: Sequence Averaging

Problem: Given a sequence $a_1, a_2, \ldots, a_{2025}$ such that $a_1 = a_{2025} = 90$ and every element is the average of its neighbors ($a_i = \frac{a_{i-1}+a_{i+1}}{2}$), prove that all numbers are equal.

Proof Sketch:

- Assume not all a_i are equal.
- Let a_k be the largest element (maximum).
- Since $a_k = \frac{a_{k-1}+a_{k+1}}{2}$, a_k is the average of its neighbors.
- This equality holds only if $a_{k-1} = a_k = a_{k+1}$ (if neighbors were smaller, the average would be smaller).
- Propagating this logic implies the neighbors of the neighbors are also equal to the maximum, and so on.
- Thus, the sequence must be constant.